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EXP NO: 1
12/01/2026

AZURE DEVOPS ENVIRONMENT SETUP

AIM:-

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

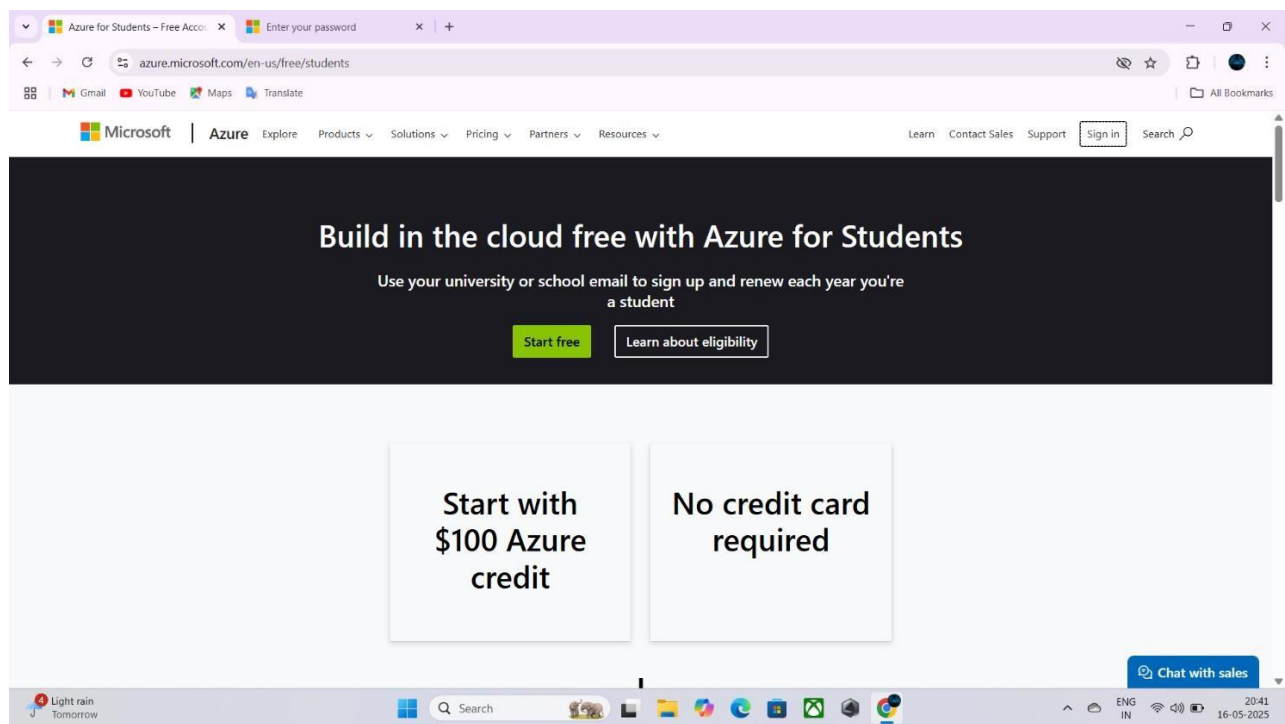
INSTALLATION:-

1. Open your web browser and go to the Azure website:

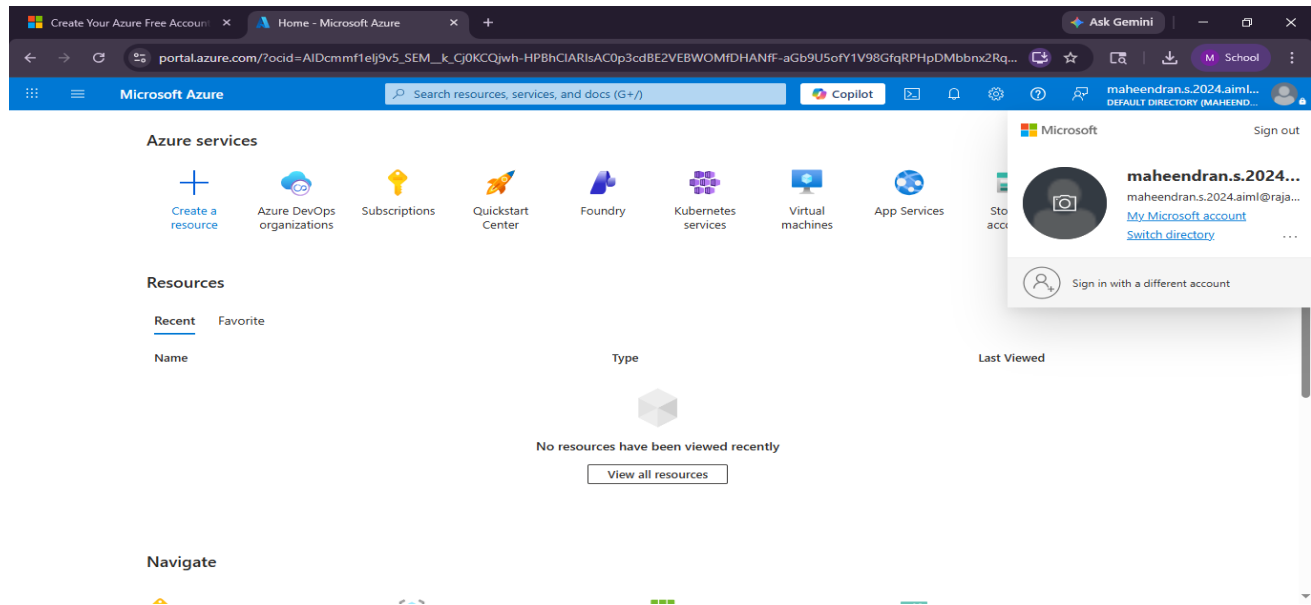
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

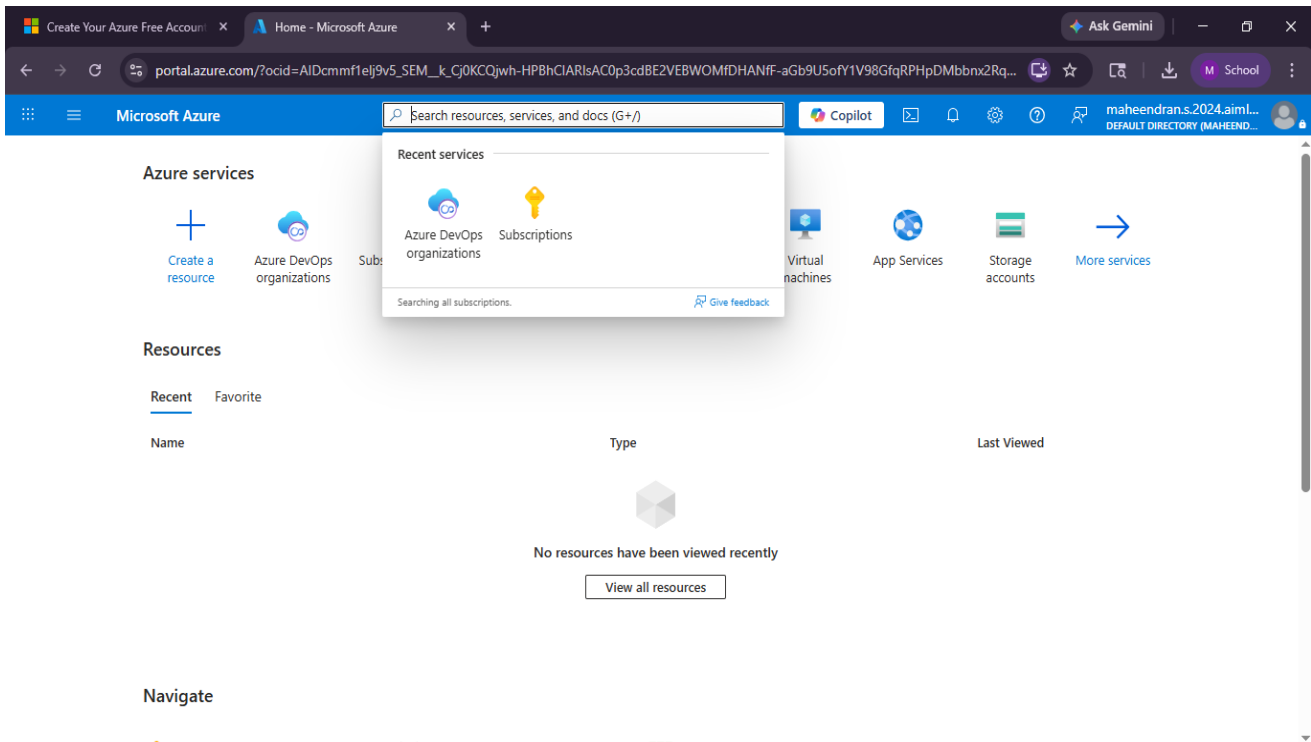
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



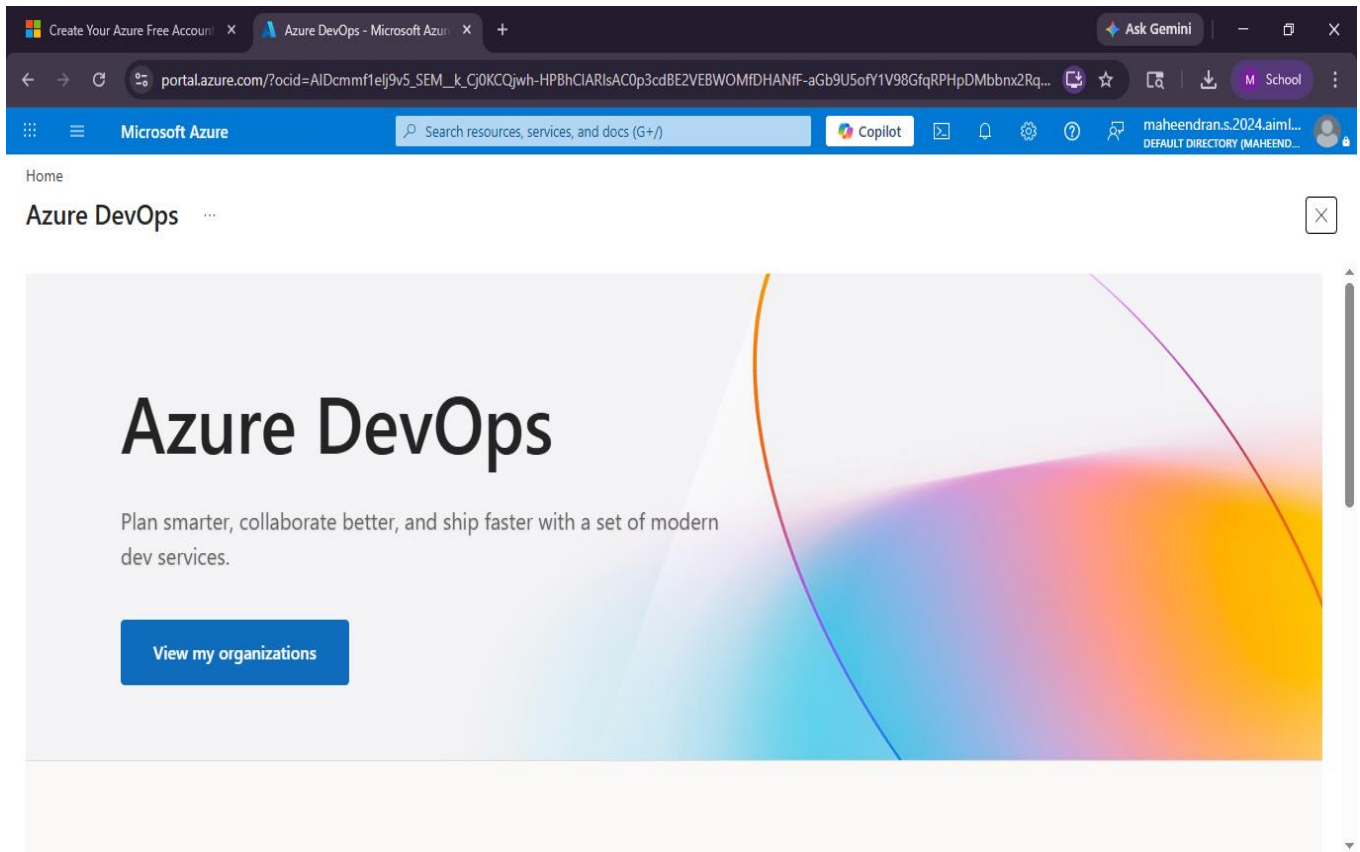
2. Azure home page



3. Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



RESULT:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

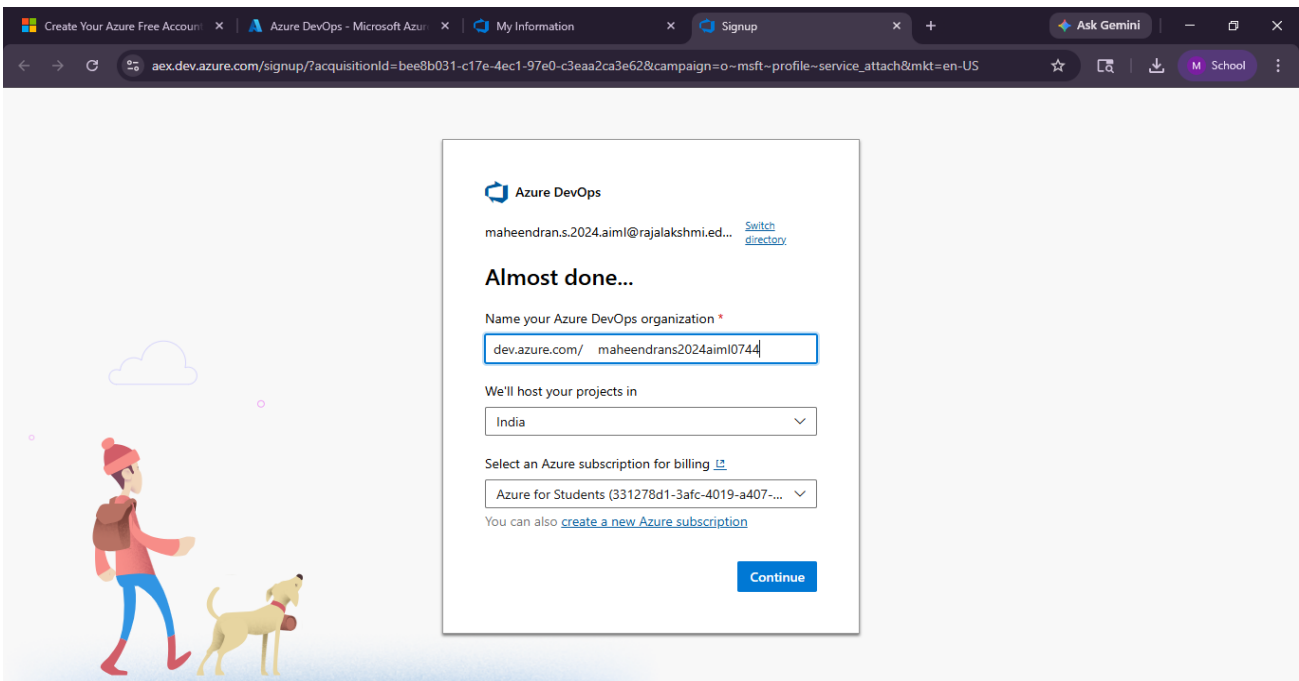
EXP NO: 2
12/01/2026

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

AIM:-

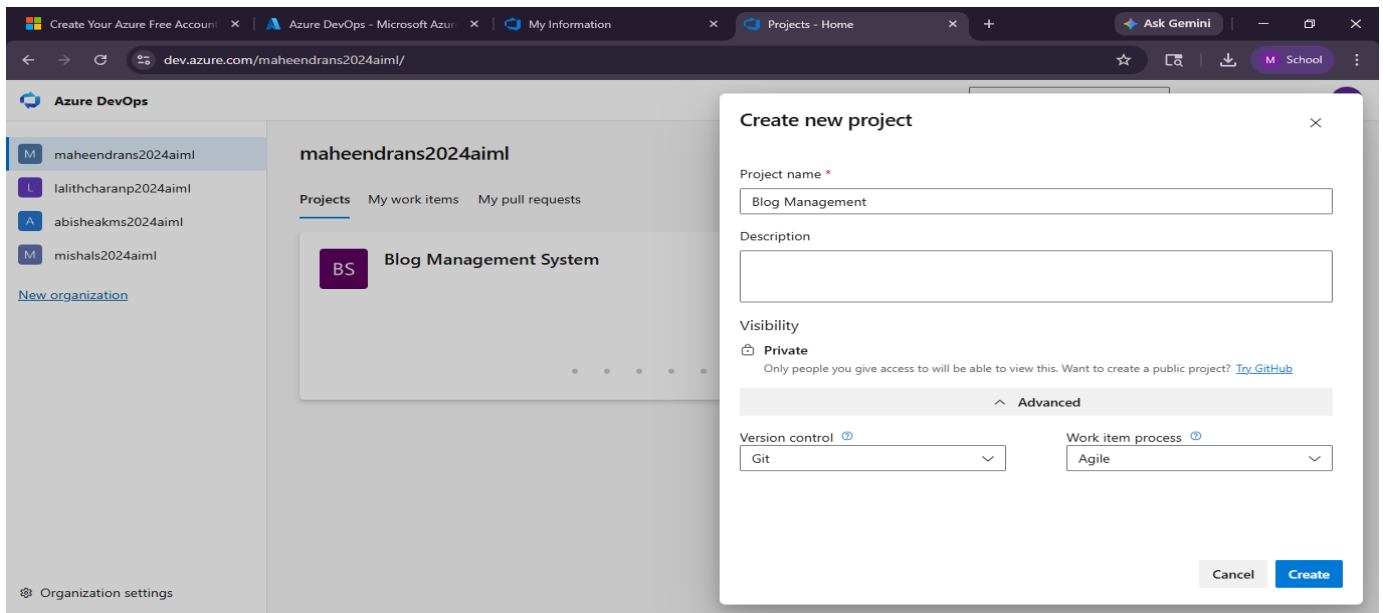
To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account

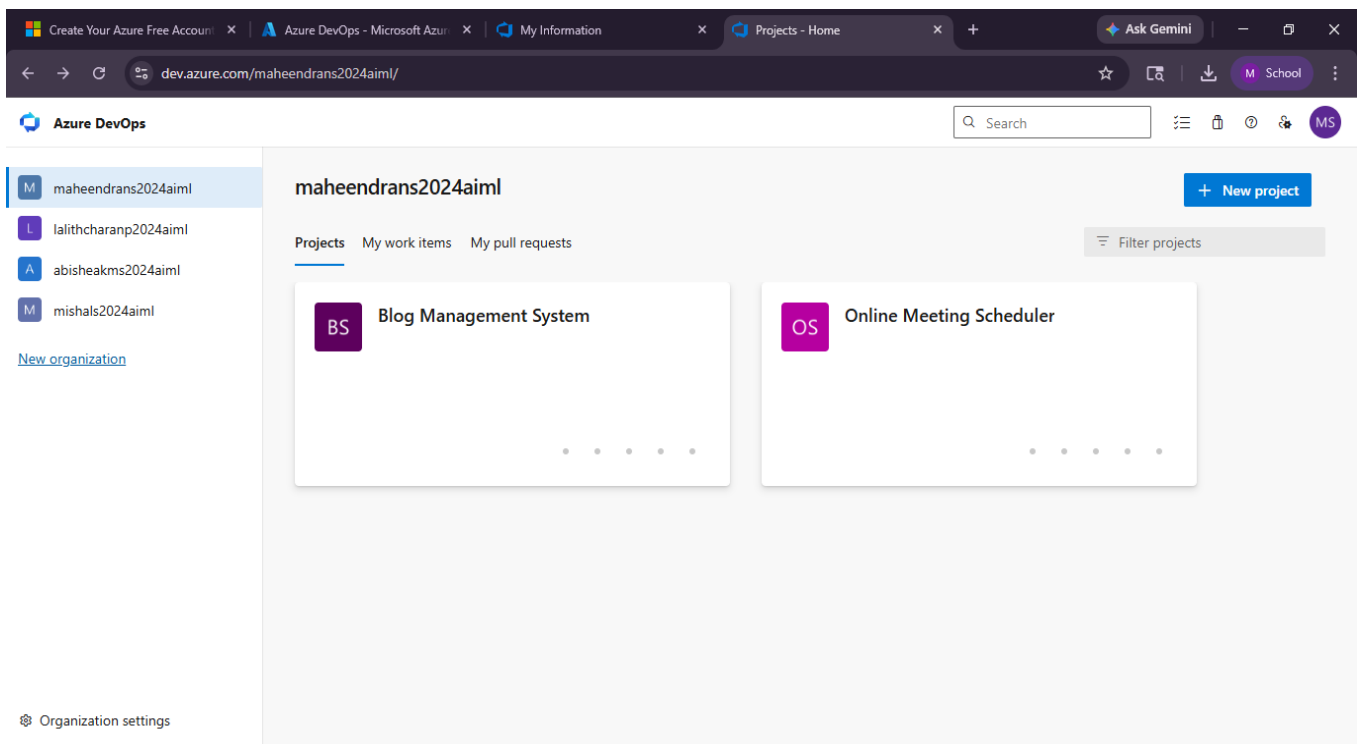


2. Create the First Project in Your Organization

- a) After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- b) On the organization's **Home page**, click on the **New Project** button.
- c) Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- d) Once you've filled out the details, click **Create** to set up your first project.



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



4. Project dashboard

The screenshot shows the Azure DevOps interface for a project named 'Blog Management System'. The browser address bar shows the URL: `dev.azure.com/maheendrans2024aiml/Blog%20Management%20System`. The page header includes the Azure DevOps logo, the project name, and navigation links for Overview, Summary, Dashboards, Wiki, Boards, Repos, Pipelines, Test Plans, and Artifacts. The main content area is titled 'About this project' and includes a prompt to 'Help others to get on board!' with a button to '+ Add Project Description'. To the right, the 'Project stats' section shows metrics for the last 7 days: 0 Work items created, 0 Work items completed, 0 Pull requests opened, and 0 Commits by 0 authors. The Pipelines section shows a 100% success rate for builds.

Azure DevOps maheendrans2024aiml / Blog Management System / Overview / Summary

Blog Management System Private Invite

About this project

Help others to get on board!

Describe your project and make it easier for other people to understand it.

+ Add Project Description

Project stats Period: Last 7 days

Boards

0 Work items created

0 Work items completed

Repos

0 Pull requests opened

0 Commits by 0 authors

Pipelines

100%

Builds succeeded

5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot displays the Azure DevOps interface for the 'Blog Management System' project. The left-hand navigation menu is open, showing various project components. The 'Work items' tab is selected, displaying a list of work items. The table below represents the data shown in the screenshot.

ID	Title	Assigned To	State	Area Path	Tags	Comments
80	Profile View Error	MISHAL S	New	Blog Management System		
79	Publish Error	MISHAL S	New	Blog Management System		
78	TC-10 Validation for empty blog post	Maheendran S	Design	Blog Management System		
77	TC-9 Logout	Maheendran S	Design	Blog Management System		
76	TC-8 Search blog posts	Maheendran S	Design	Blog Management System		
75	TC-7 Add comment to blog	Maheendran S	Design	Blog Management System		
74	TC-6 View blog post	Maheendran S	Design	Blog Management System		
73	TC-5 Delete blog post	Maheendran S	Design	Blog Management System		
72	TC-4 Edit blog post	Maheendran S	Design	Blog Management System		
71	TC-3 Create new blog post	Maheendran S	Design	Blog Management System		
70	TC-2 User Login with invalid credentials	Maheendran S	Design	Blog Management System		
69	TC-1 User Login with valid credentials	Maheendran S	Design	Blog Management System		
34	As an admin, I want to delete unused categories so that system stays	Unassigned	New	Blog Management System		

Azure DevOps maheendrans2024aiml / Blog Management System / Boards / Work items

Search

Work items

Recently updated ▾ + New Work Item ▾ Open in Queries Column Options Import Work Items Recycle Bin

Filter by keyword Types Assigned to

ID	Title	Assigned To	State	Area Path
69	TC-1 User Login with valid credentials	Maheendran S	Design	Blog Management System
34	As an admin, I want to delete unused categories so that system stays	Unassigned	New	Blog Management System
33	As an admin, I want to edit categories so that I can update classificat	Unassigned	New	Blog Management System
32	As an admin, I want to create blog categories so that posts are organi	Unassigned	New	Blog Management System
29	As an admin, I want to block users who violate guidelines so that mis	Unassigned	New	Blog Management System
28	As an admin, I want to delete inappropriate comments so that the pla	Unassigned	New	Blog Management System
27	As an admin, I want to approve comments before publishing so that s	Unassigned	New	Blog Management System
25	As a reader, I want to delete my comment so that I can remove unwar	Unassigned	New	Blog Management System
24	As a reader, I want to edit my comment so that I can correct mistakes	Unassigned	New	Blog Management System
23	As a reader, I want to comment on blog posts so that I can share my c	Unassigned	New	Blog Management System
20	As an author, I want to schedule publishing so that posts go live auto	Unassigned	New	Blog Management System
19	As an author, I want to publish my blog post so that readers can view	Unassigned	New	Blog Management System
18	As an author, I want to delete my blog post so that I can remove unw	Unassigned	New	Blog Management System
16	As an author, I want to save a draft so that I can publish later.	Unassigned	New	Blog Management System

Microsoft Sign out

Maheendran S
maheendrans.2024.aiml@raja...
[My Microsoft account](#)
[Switch directory](#)

Sign in with a different account

Result

Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3
19/01/2026

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for a project named 'Blog Management System Team'. The 'Backlog' view is active, displaying a list of work items. The left sidebar shows navigation options like Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Planning Poker, and Repos. The main area shows a table of work items with columns for Order, Work Item Type, Title, and State. The items are organized into five epics: User Management, User Registration, User Authentication, Blog Management, and Comment Management. The right sidebar shows a 'Planning' panel with a 'Blog Management System Team backlog' and three sprints: Sprint 1 (2/2/2026 - 2/16/2026), Sprint 2 (2/17/2026 - 3/3/2026), and Sprint 3 (3/4/2026 - 3/18/2026).

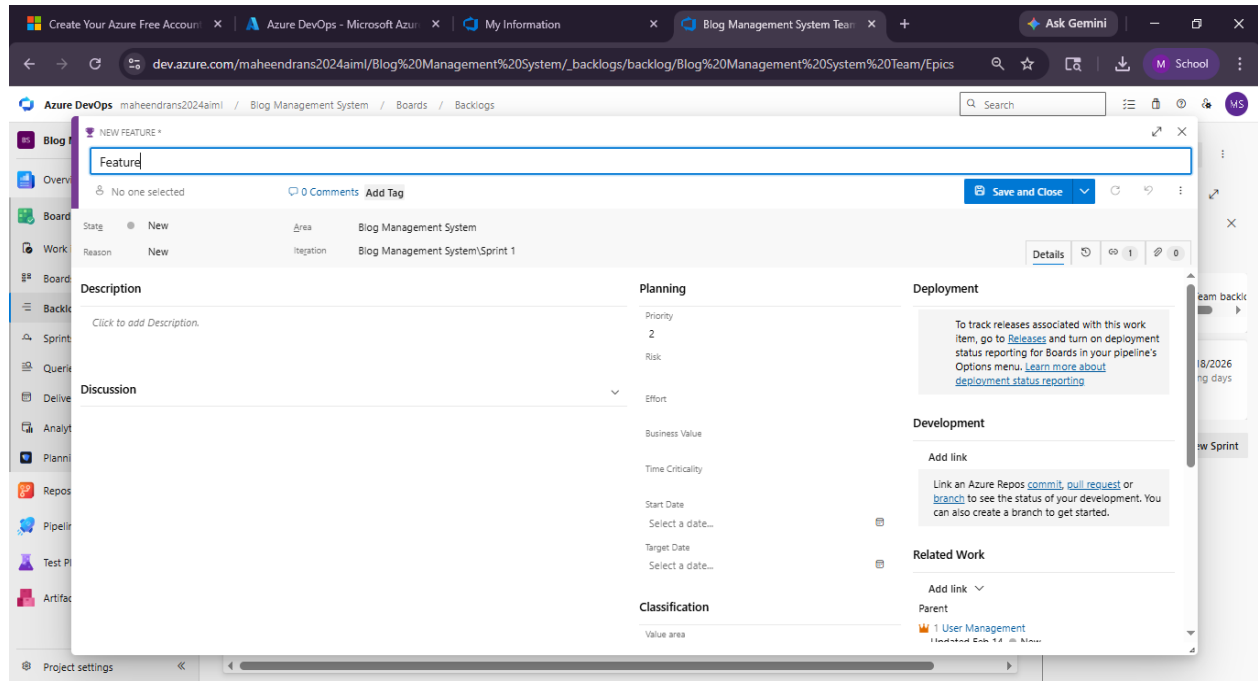
Order	Work Item Type	Title	State
1	Epic	User Management	New
	Feature	User Registration	New
	User Story	As a new user, I want to register with email and pass...	New
	User Story	As a user, I want to receive email verification so that m...	New
	User Story	As a user, I want to see validation messages for incor...	New
	Feature	User Authentication	New
2	Epic	Blog Management	New
3	Epic	Comment Management	New
4	Epic	Category & Tag Management	New
5	Epic	Search & Analytics	New

1.Fill in Epics

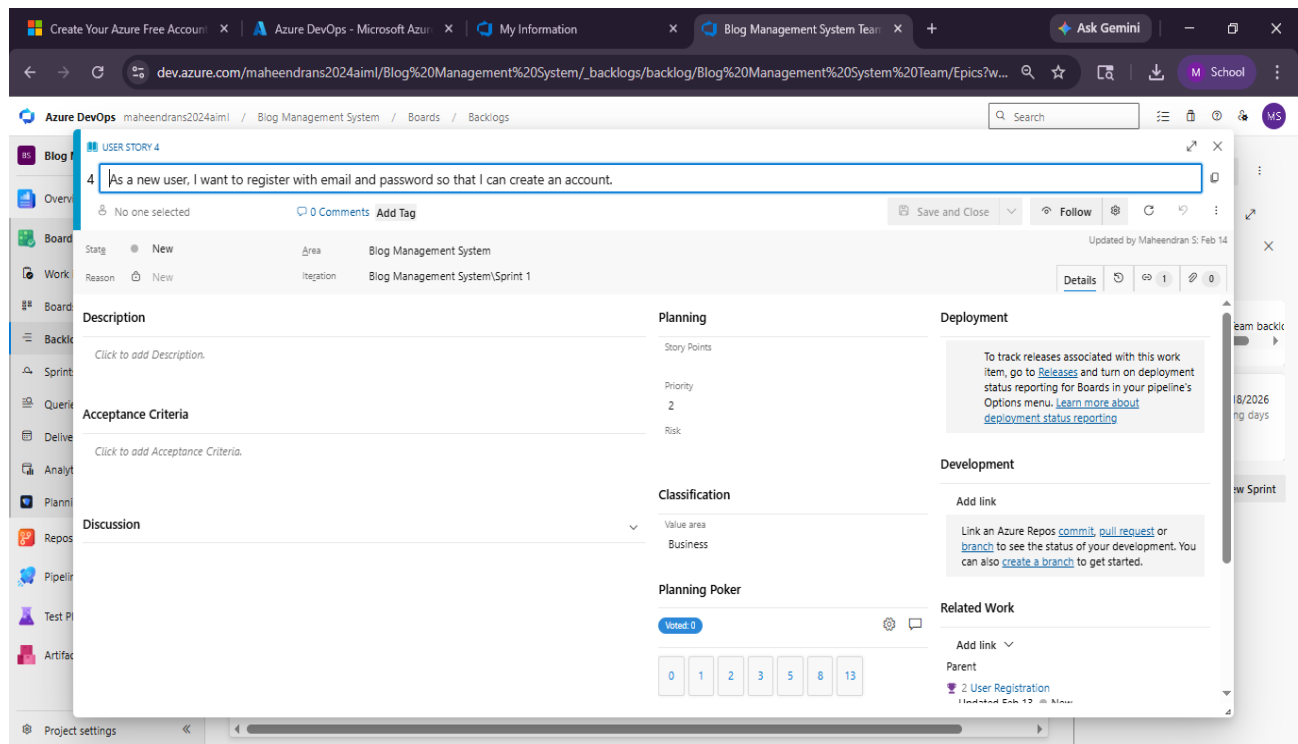
This screenshot shows the same Azure DevOps interface, but with the 'Epics' filter applied to the backlog. The table now includes additional columns: Effort, Business Area, and Tags. The work items are filtered to show only those belonging to the 'Business' area. The 'Planning' panel on the right is also visible, showing the same sprint information as the first screenshot.

Order	Work Item Type	Title	State	Effort	Business Area	Tags
1	Epic	User Management	New		Business	
	Feature	User Registration	New		Business	
	User Story	As a new user, I want to register with email and pass...	New		Business	
	User Story	As a user, I want to receive email verification so that ...	New		Business	
	User Story	As a user, I want to see validation messages for incor...	New		Business	
	Feature	User Authentication	New		Business	
2	Epic	Blog Management	New		Business	
	Feature	Create & Edit Blog	New		Business	
	Feature	Delete & Publish Blog	New		Business	
3	Epic	Comment Management	New		Business	
	Feature	Add Comments	New		Business	
	Feature	Moderate Comments	New		Business	
4	Epic	Category & Tag Management	New		Business	
5	Epic	Search & Analytics	New		Business	

2.Fill in Features



3.Fill in User Story Details



RESULT:-

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4

19/01/2026

**LIST OF FUNCTIONAL AND NON FUNCTIONAL
REQUIREMENTS**

Aim

To design and develop a **Blog Management System** that allows users to create, edit, publish, and manage blog posts efficiently, while providing administrators control over content, users, and system operations.

Algorithm

1. **Start**
2. User opens the system
3. Display homepage with available blogs
4. If user selects **Register**
 - Input user details
 - Validate data
 - Store in database
5. If user selects **Login**
 - Enter username & password
 - Validate credentials
 - Grant access
6. After login:
 - Display dashboard
7. User actions:
 - Create new blog → Input content → Save → Publish
 - Edit blog → Update → Save
 - Delete blog → Confirm → Remove
8. Admin actions (if admin login):
 - Manage users
 - Approve/reject blogs
 - Monitor system activity
9. Display updated blog list
10. Logout
11. **End**

Functional Requirements

1. The system shall allow users to register by providing necessary details.
2. The system shall allow registered users to log in using valid credentials.
3. The system shall allow users to create new blog posts.
4. The system shall allow users to edit their existing blog posts.
5. The system shall allow users to delete their blog posts.
6. The system shall allow users to view all published blog posts.
7. The system shall allow users to search and filter blog posts.
8. The system shall allow users to add comments to blog posts.

Non-Functional Requirements

1. **Performance** – System should load pages within 2–3 seconds
2. **Scalability** – Should handle increasing users and blog data
3. **Security** – Protect user data with authentication and encryption
4. **Usability** – User-friendly interface for all age groups
5. **Reliability** – System should be available with minimal downtime
6. **Maintainability** – Easy to update and fix issues
7. **Compatibility** – Works across browsers and devices
8. **Backup & Recovery** – Regular data backup and restore mechanisms

Result

The Blog Management System is successfully developed to allow users to create and manage blogs efficiently. It provides a secure, scalable, and user-friendly platform for content publishing and administration, fulfilling all functional and non-functional requirements.

EXP NO: 5

02/02/2026

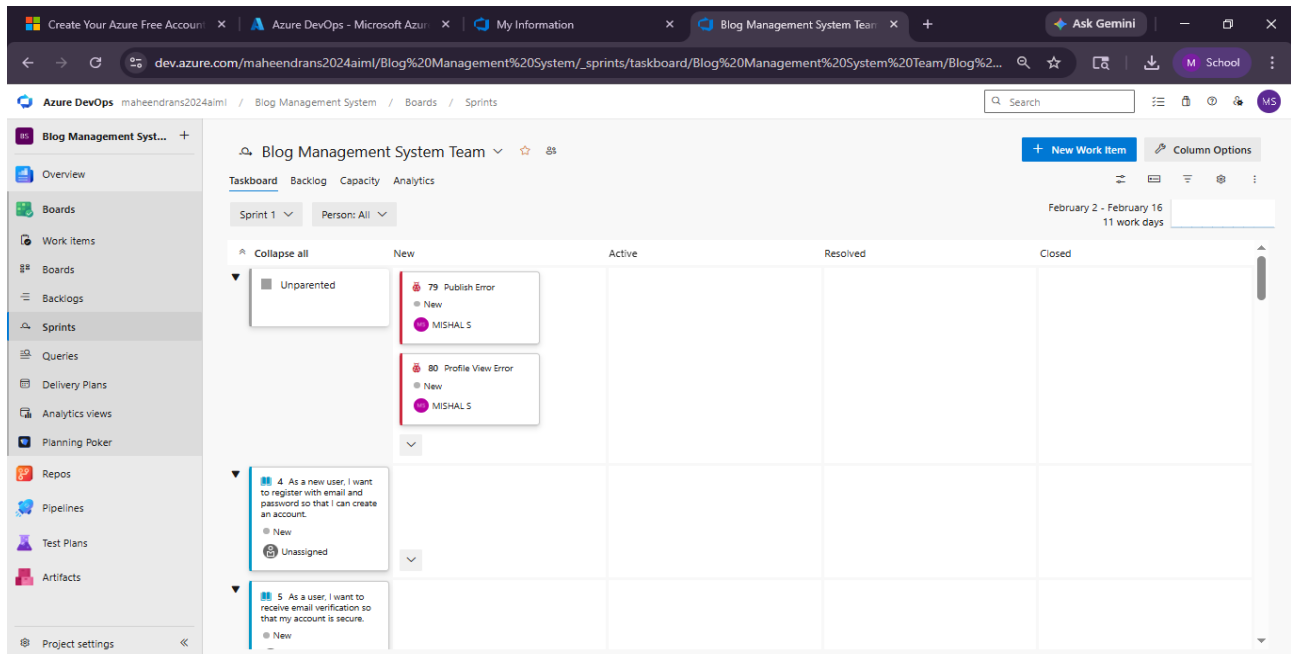
SPRINT PLANNING

AIM:-

To assign user story to specific sprint for the Blog Management System Project.

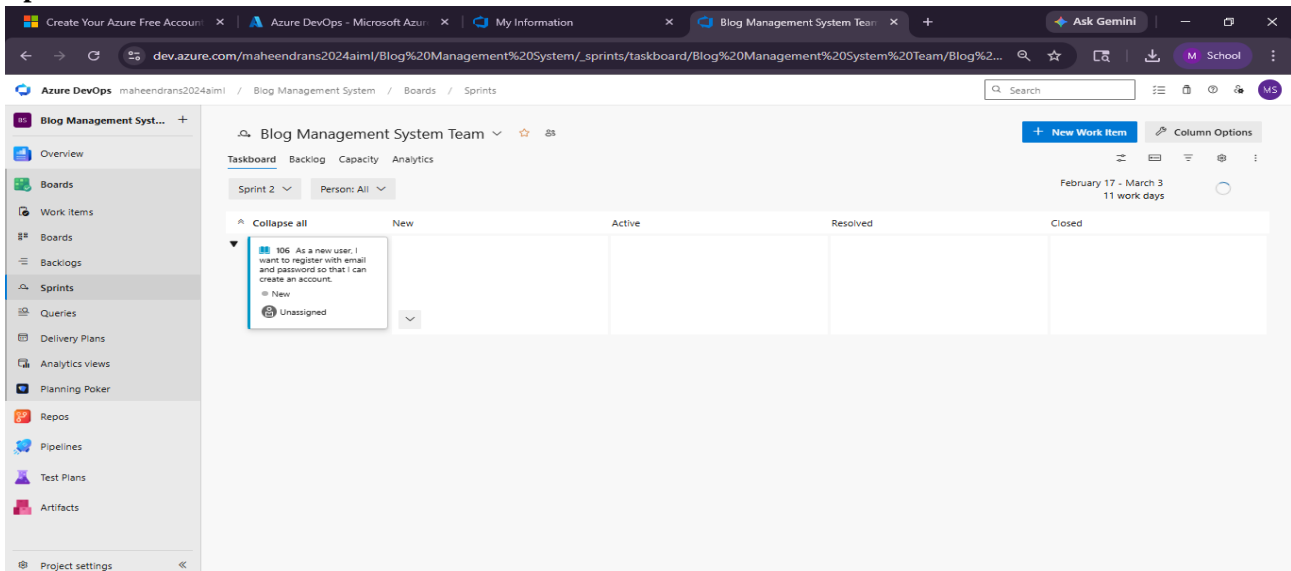
Sprint Planning:-

Sprint 1



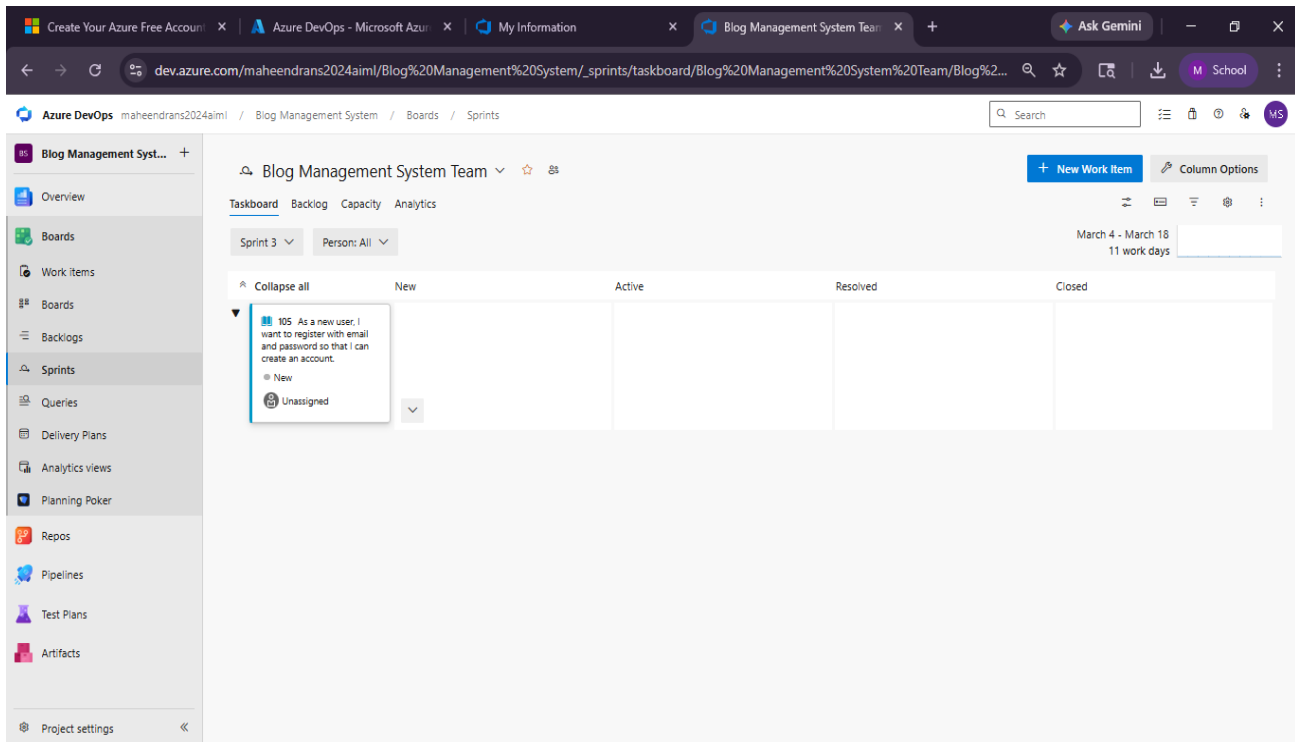
The screenshot shows the Azure DevOps interface for the 'Blog Management System Team'. The 'Sprints' section is active, displaying a Kanban board for 'Sprint 1' (February 2 - February 16, 11 work days). The board has columns for 'New', 'Active', 'Resolved', and 'Closed'. Under the 'New' column, there are three items: 'Unparented', '79 Publish Error' (New, MISHAL S), and '80 Profile View Error' (New, MISHAL S). Below these, there are two user stories: '4 As a new user, I want to register with email and password so that I can create an account.' (New, Unassigned) and '5 As a user, I want to receive email verification so that my account is secure.' (New, Unassigned).

Sprint 2



The screenshot shows the Azure DevOps interface for the 'Blog Management System Team'. The 'Sprints' section is active, displaying a Kanban board for 'Sprint 2' (February 17 - March 3, 11 work days). The board has columns for 'New', 'Active', 'Resolved', and 'Closed'. Under the 'New' column, there is one item: '106 As a new user, I want to register with email and password so that I can create an account.' (New, Unassigned).

Sprint 3



RESULT:-

The Sprints are created for the Student Management System.

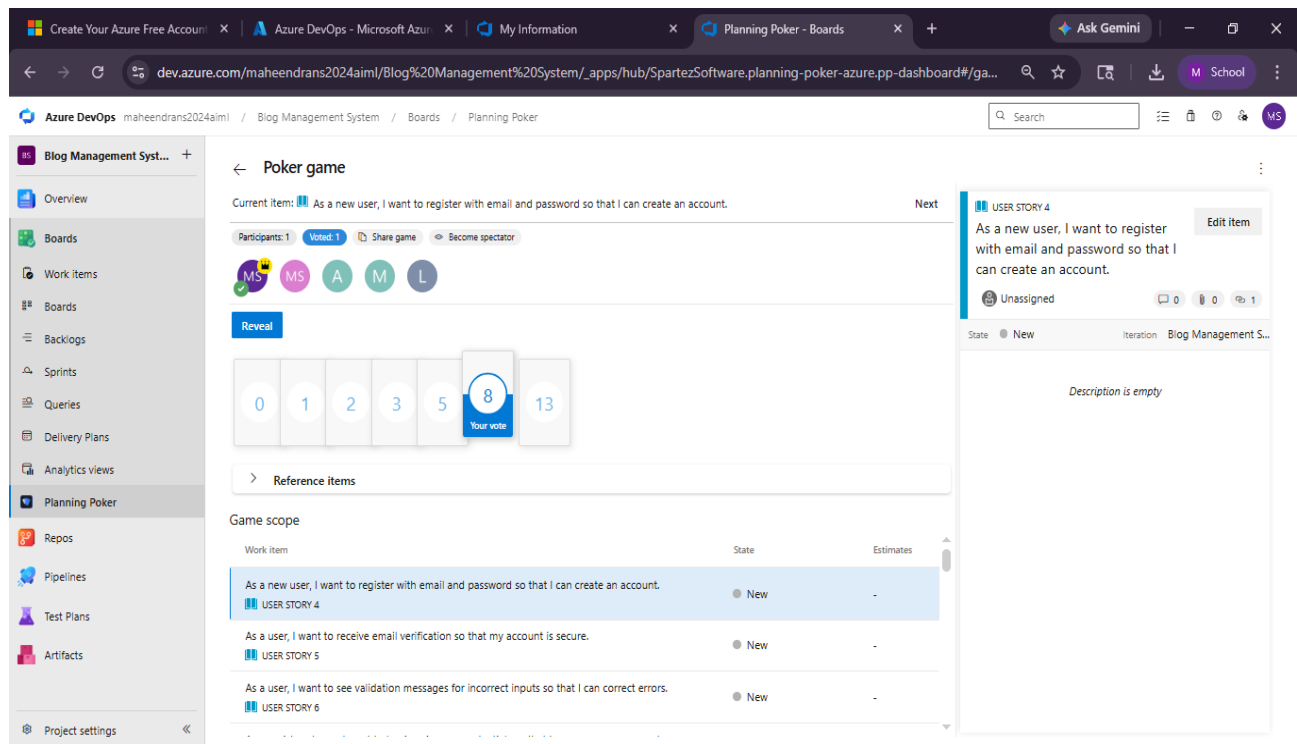
EXP NO: 6
09/02/2026

POKER ESTIMATION

AIM:-

Create Poker Estimation for the user stories -Student management System.

Poker Estimation:-



RESULT:-

The Estimation/Story Points is created for the project using Poker Estimation.

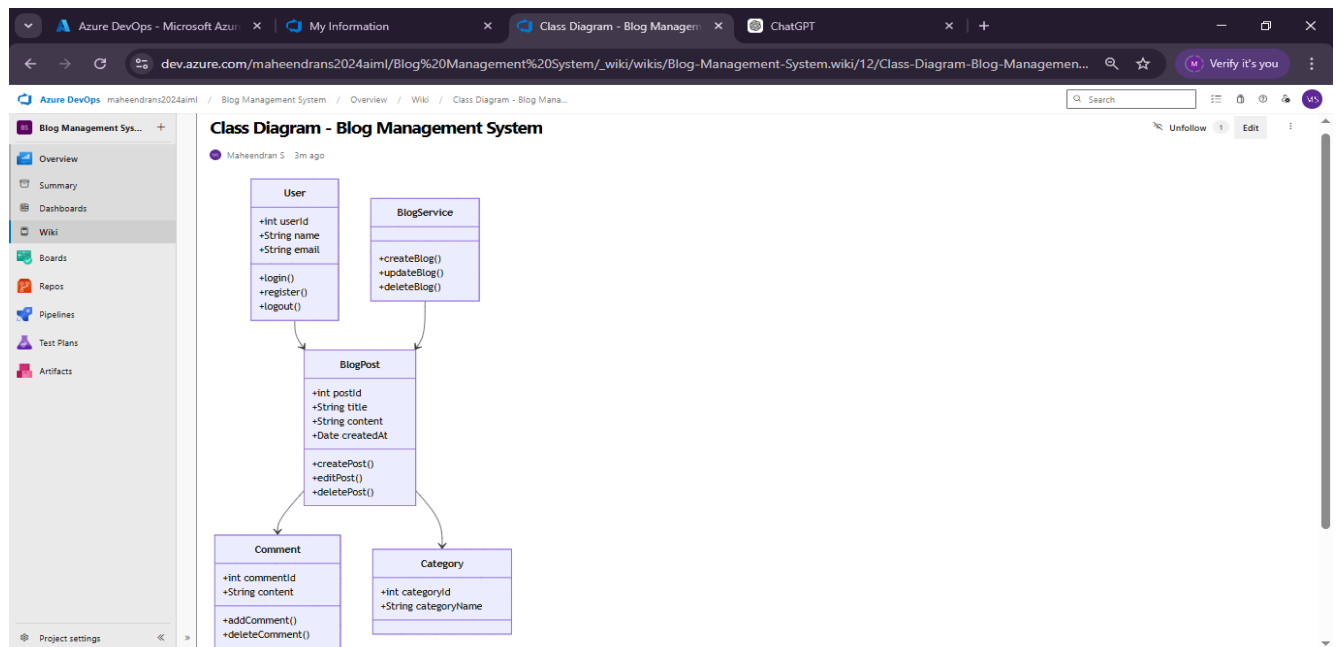
EXP NO: 7
16/02/2026

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

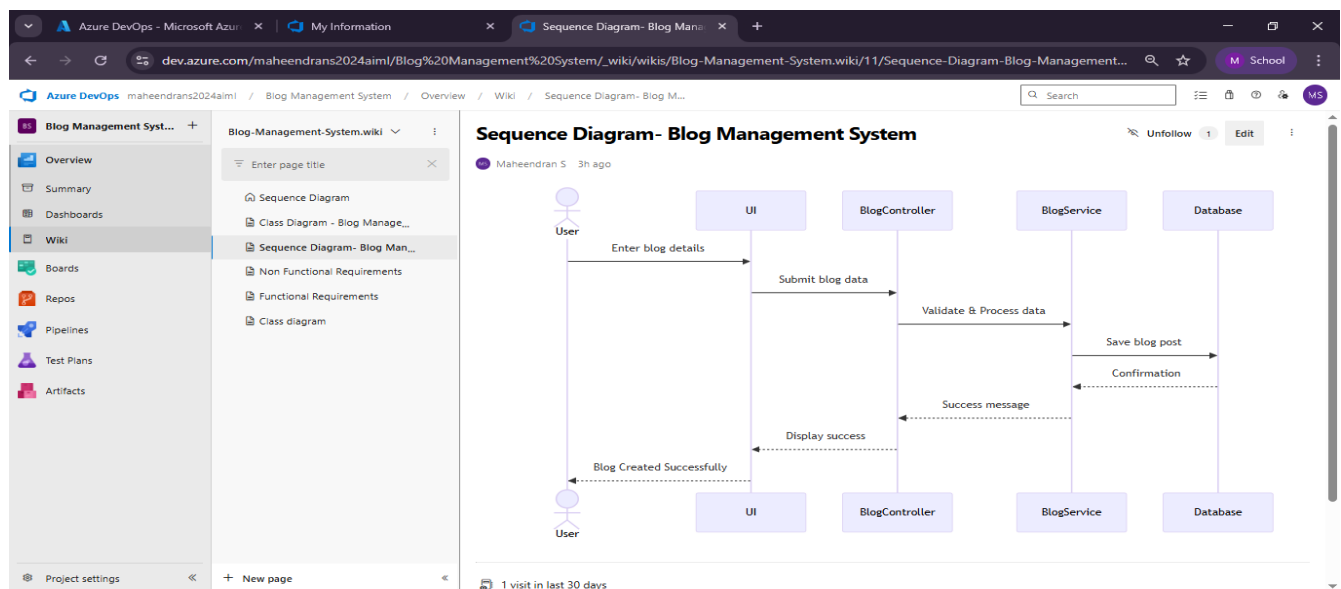
AIM:-

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. CLASS DIAGRAM:-



6B. SEQUENCE DIAGRAM:-



Result:

The Class Diagram and Sequence Diagram is designed Successfully for given project.

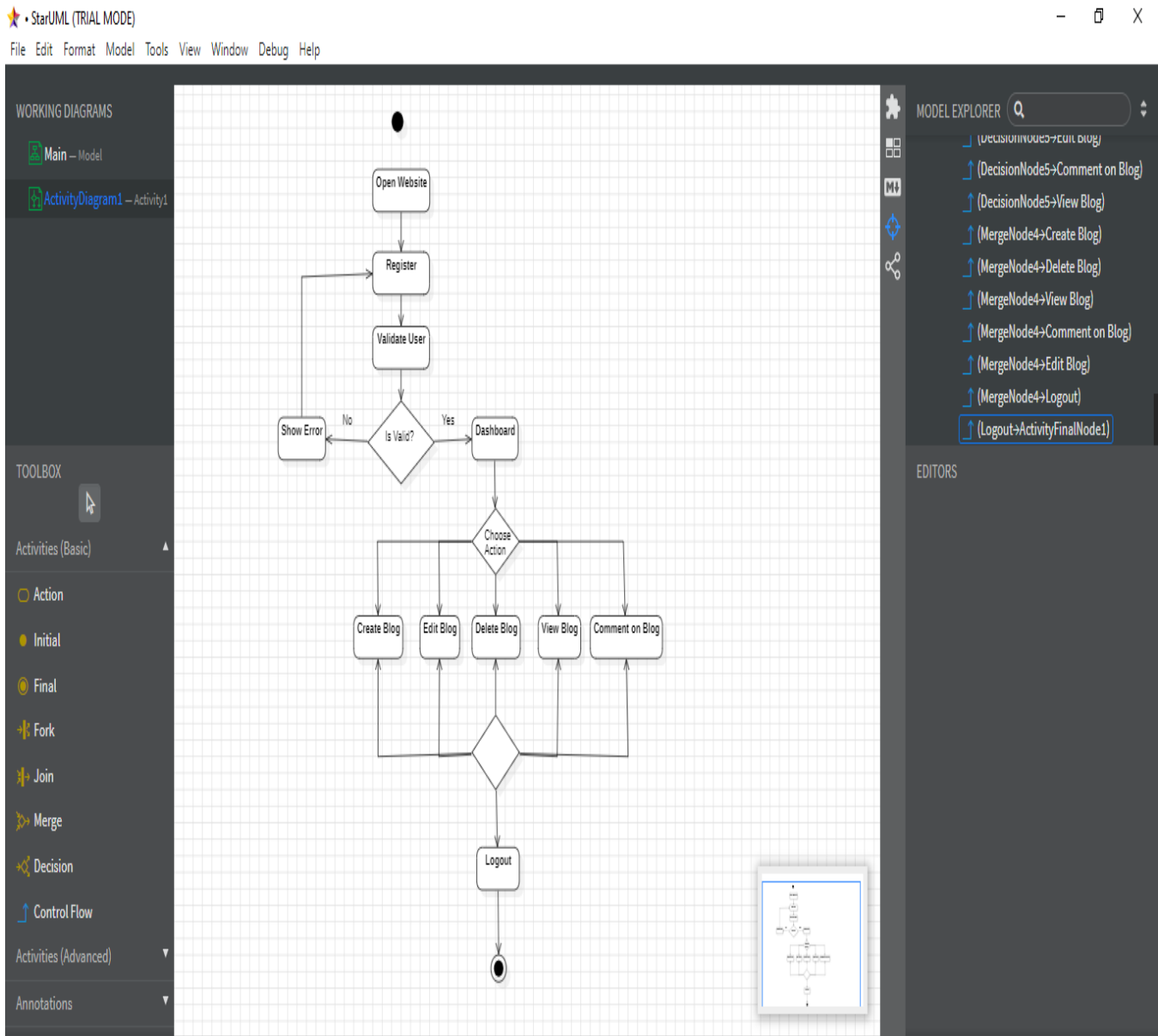
EXP NO: 8
23/02/2026

DESIGNING USE CASE DIAGRAMS AND ACTIVITY DIAGRAM FOR PROJECT ARCHITECTURE

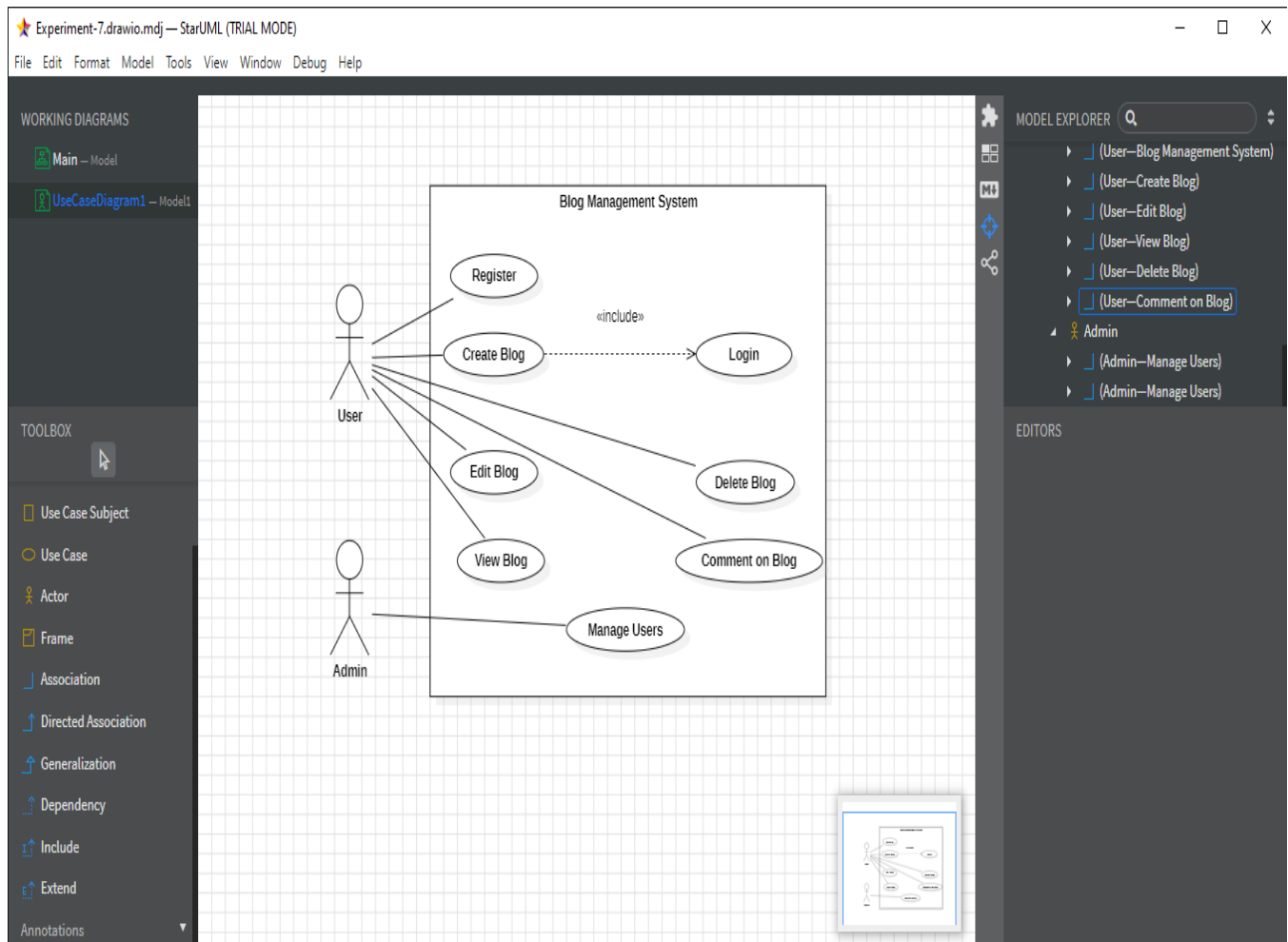
AIM:-

To Design an activity Diagram and use case Diagram for the given Project.

7A. Activity Diagram:-



7B. USE CASE DIAGRAM:-



RESULT:-

Thus activity and use case diagram has been designed successfully for Student Management System

AIM:-

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

**Test Planning and Test Case
Test Case Design Procedure****Understand Core Features of the Application**

- User Signup & Login
- Create, Edit, and Delete Blog Posts
- Commenting on Blog Posts
- Like/Share Functionality
- Viewing Blog Posts (Public/Private)
- Admin Role Management and Content Moderation

2. Define User Interactions

- Each test case simulates real user behavior (e.g., logging in, creating a post, commenting, liking a post, moderating content).

3. Design Happy Path Test Cases

- Focus on validating that all core functionalities work correctly under normal conditions.
- Example: User registers → logs in → creates a blog post → publishes it → other users view, like, and comment on the post.

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.
- Examples:
 - Login with incorrect credentials
 - Submitting an empty blog post
 - Comment submission without text
 - Unauthorized user trying to edit/delete another user's post
 - Accessing admin panel without permission

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can easily execute and validate outcomes.

6. Use Clear Naming and IDs

- Test cases are uniquely identified (e.g., TC01 – User Login Success, TC08 – Create Blog Post, TC15 – Invalid Comment Submission).
- Helps in tracking and mapping to features in tools like Azure DevOps.

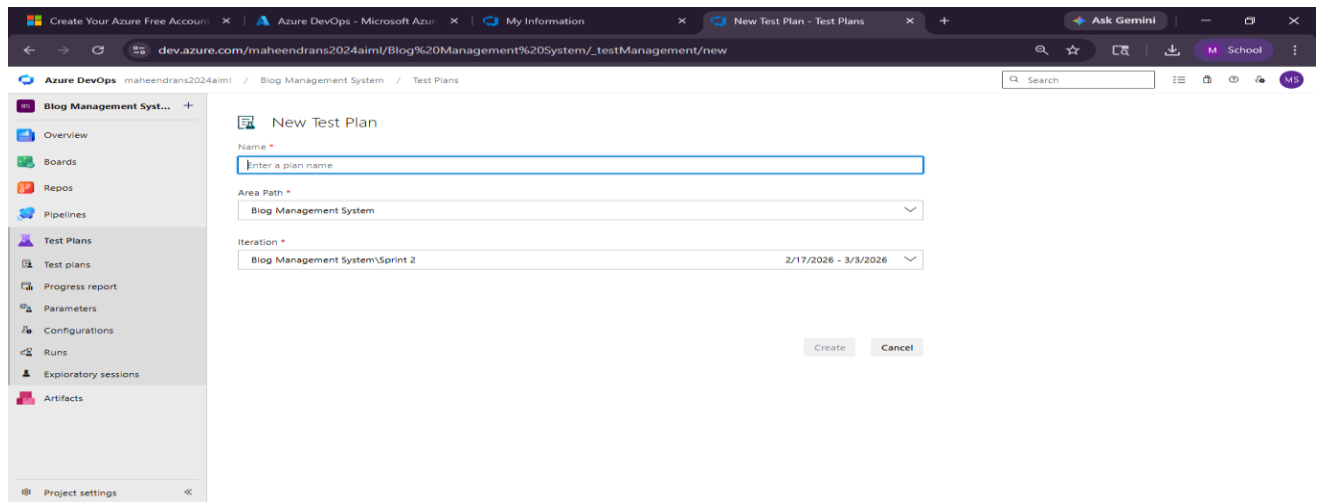
7. Separate Test Suites

- Group test cases by functionality such as:
 - Login and Registration
 - Blog Post Management
 - Comments and Interactions
 - Content Moderation
 - Admin Functions
- Improves organization and allows focused execution.

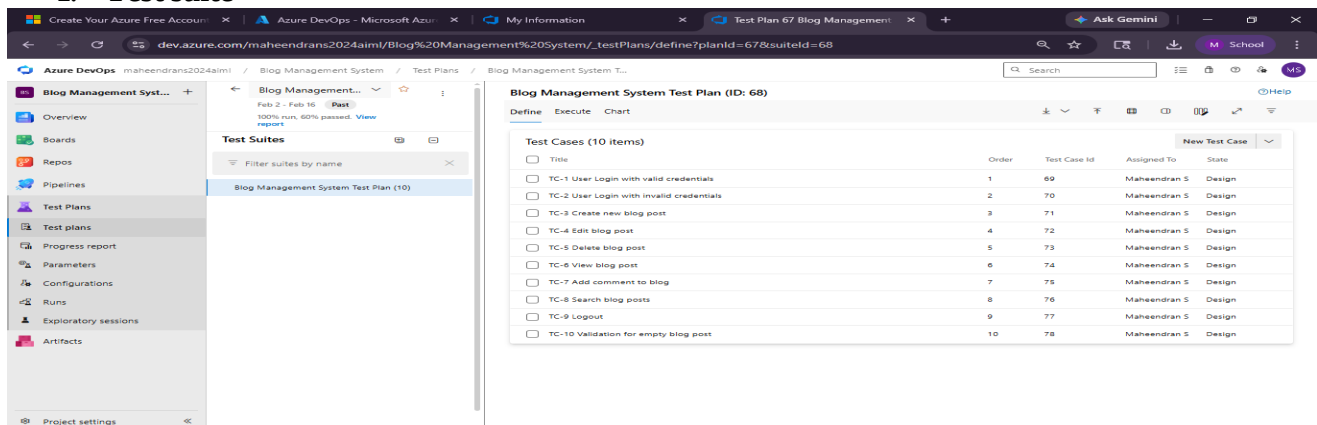
8. Prioritize and Review

- Assign high priority to critical workflows like login, post creation, publishing, and content moderation.
- Review for completeness, accuracy, and alignment with user stories and feature requirements.

1. New test plan



2. Test suite



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Student Management System– Test Plans

USER STORIES

- **ID: 201** – As a user, I want to log in securely so I can access my blog dashboard
- **ID: 202** – As a user, I want to create blog posts so I can share content
- **ID: 203** – As a user, I want to edit blog posts so I can update content
- **ID: 204** – As a user, I want to delete blog posts so I can remove unwanted content
- **ID: 205** – As a user, I want to search blog posts to find specific content

Test Suite: TS01 – User Authentication

- **TC-1: User login with valid credentials**
 - User Story: ID 201
 - Type: Happy Path
- **TC-2: User login with invalid credentials**
 - User Story: ID 201
 - Type: Error Path

Test Suite: TS02 – Blog Creation

- **TC-3: Create new blog post with valid data**
 - User Story: ID 202
 - Type: Happy Path
- **TC-4: Create blog post with empty required fields**
 - User Story: ID 202
 - Type: Error Path

Test Suite: TS03 – Blog Editing

- **TC-5: Edit existing blog post successfully**
 - User Story: ID 203
 - Type: Happy Path
- **TC-6: Edit blog post with invalid or empty content**
 - User Story: ID 203
 - Type: Error Path

Test Suite: TS04 – Blog Deletion

- **TC-7: Delete blog post successfully**
 - User Story: ID 204
 - Type: Happy Path
- **TC-8: Attempt to delete non-existing blog post**
 - User Story: ID 204
 - Type: Error Path

Test Suite: TS05 – Blog Search

- **TC-9: Search blog posts with valid keyword**
 - User Story: ID 205
 - Type: Happy Path
- **TC-10: Search blog posts with no matching results**
 - User Story: ID 205
 - Type: Error Path

1. Test Cases

The screenshot shows the Azure DevOps Test Plan interface for the 'Blog Management System Test Plan (ID: 68)'. The left sidebar contains a navigation menu with options like Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main area displays the test plan details, including a 'Test Suites' section with a filter and a 'Test Cases (10 items)' table.

Title	Order	Test Case Id	Assigned To	State
TC-1 User Login with valid credentials	1	69	Maheendran S	Design
TC-2 User Login with invalid credentials	2	70	Maheendran S	Design
TC-3 Create new blog post	3	71	Maheendran S	Design
TC-4 Edit blog post	4	72	Maheendran S	Design
TC-5 Delete blog post	5	73	Maheendran S	Design
TC-6 View blog post	6	74	Maheendran S	Design
TC-7 Add comment to blog	7	75	Maheendran S	Design
TC-8 Search blog posts	8	76	Maheendran S	Design
TC-9 Logout	9	77	Maheendran S	Design
TC-10 Validation for empty blog post	10	78	Maheendran S	Design

The screenshot shows the details of a specific test case, 'TC-1 User Login with valid credentials' (ID: 69), assigned to Maheendran S. The page includes a 'Steps' section with a table of actions and expected results, a 'Recent test results' section showing a passed status, and a 'Deployment' section with instructions on how to track releases.

Steps	Action	Expected result	Attachments
1.	Enter valid username	User logs in successfully	
2.	Enter password	User logs in successfully	
3.	Click Login	User logs in successfully	

Recent test results

Passed by Maheendran S with Windows 10
Completed 5:01 AM, 4/5/2026

Deployment

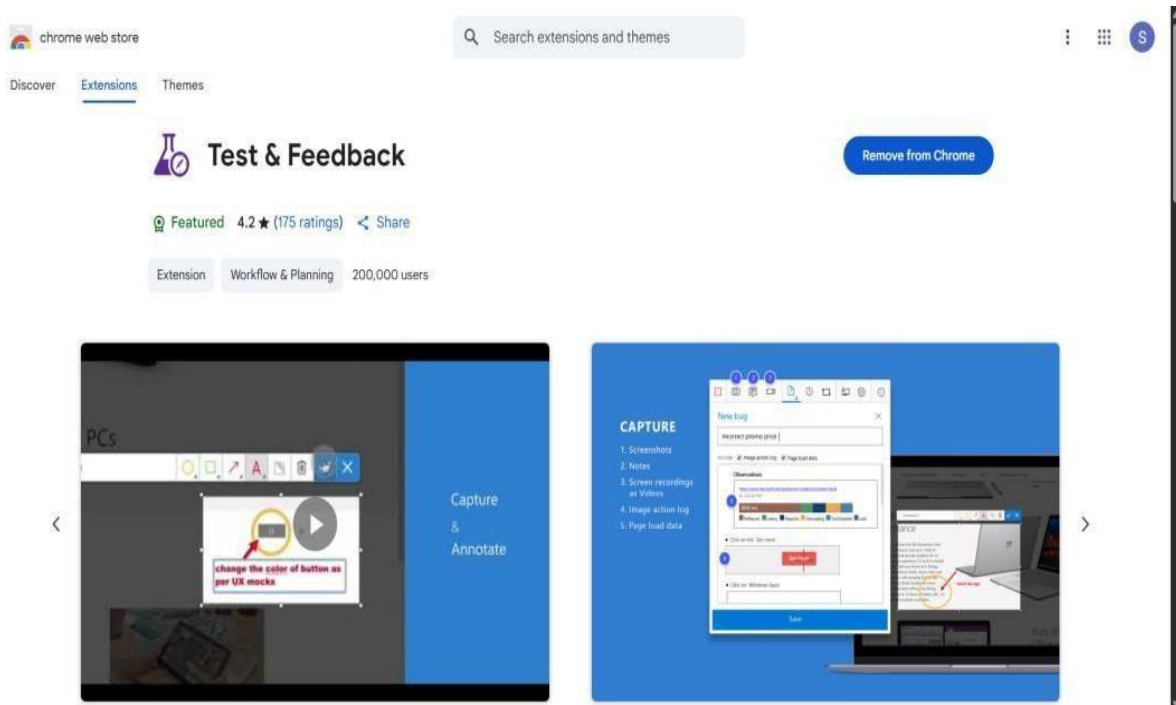
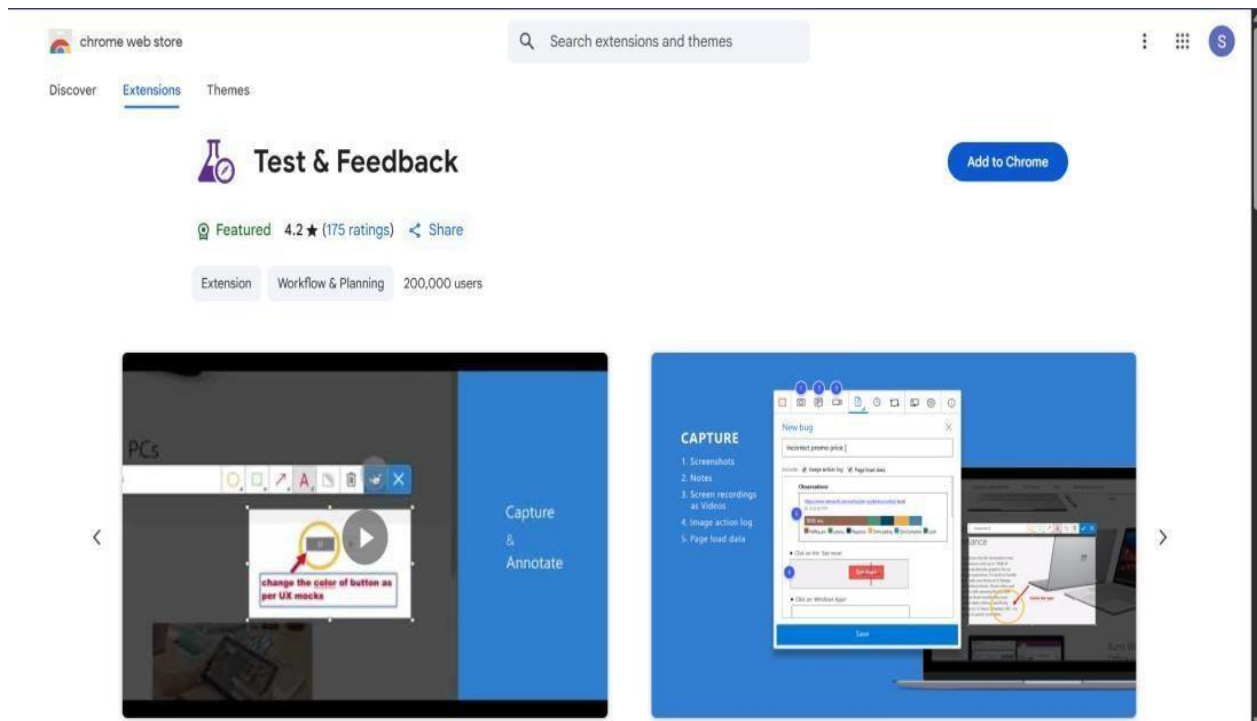
To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

2. Installation of test



3. Running the test cases

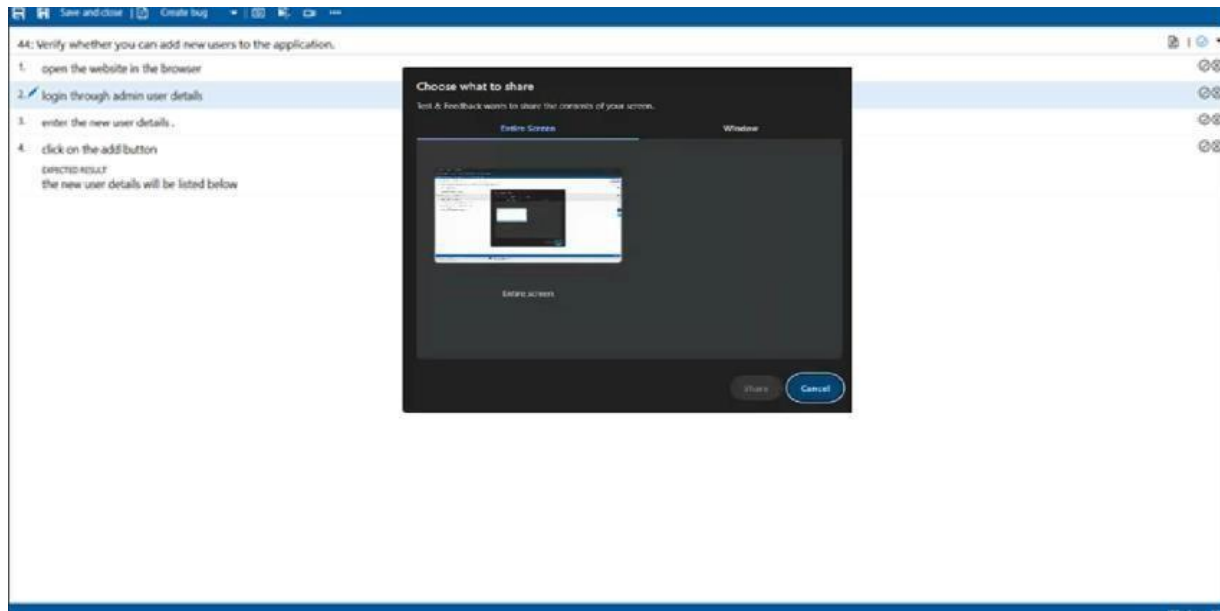
The screenshot shows the Azure DevOps Test Plan interface for a project named 'Blog Management System'. The left sidebar contains navigation options: Overview, Boards, Repos, Pipelines, Test Plans (selected), Progress report, Parameters, Configurations, Runs, and Exploratory sessions. The main area displays the 'Blog Management System Test Plan (ID: 68)' with tabs for Define, Execute, and Chart. The 'Execute' tab shows a table of test points with their outcomes.

Test Points (10 items)	Outcome
☐ Title	
☐ TC-1 User Login with valid credentials	Passed
☐ TC-2 User Login with invalid credentials	Failed
☐ TC-3 Create new blog post	Failed
☐ TC-4 Edit blog post	Passed
☐ TC-5 Delete blog post	Passed
☐ TC-6 View blog post	Failed
☐ TC-7 Add comment to blog	Passed
☐ TC-8 Search blog posts	Passed
☐ TC-9 Logout	Failed
☐ TC-10 Validation for empty blog post	Passed

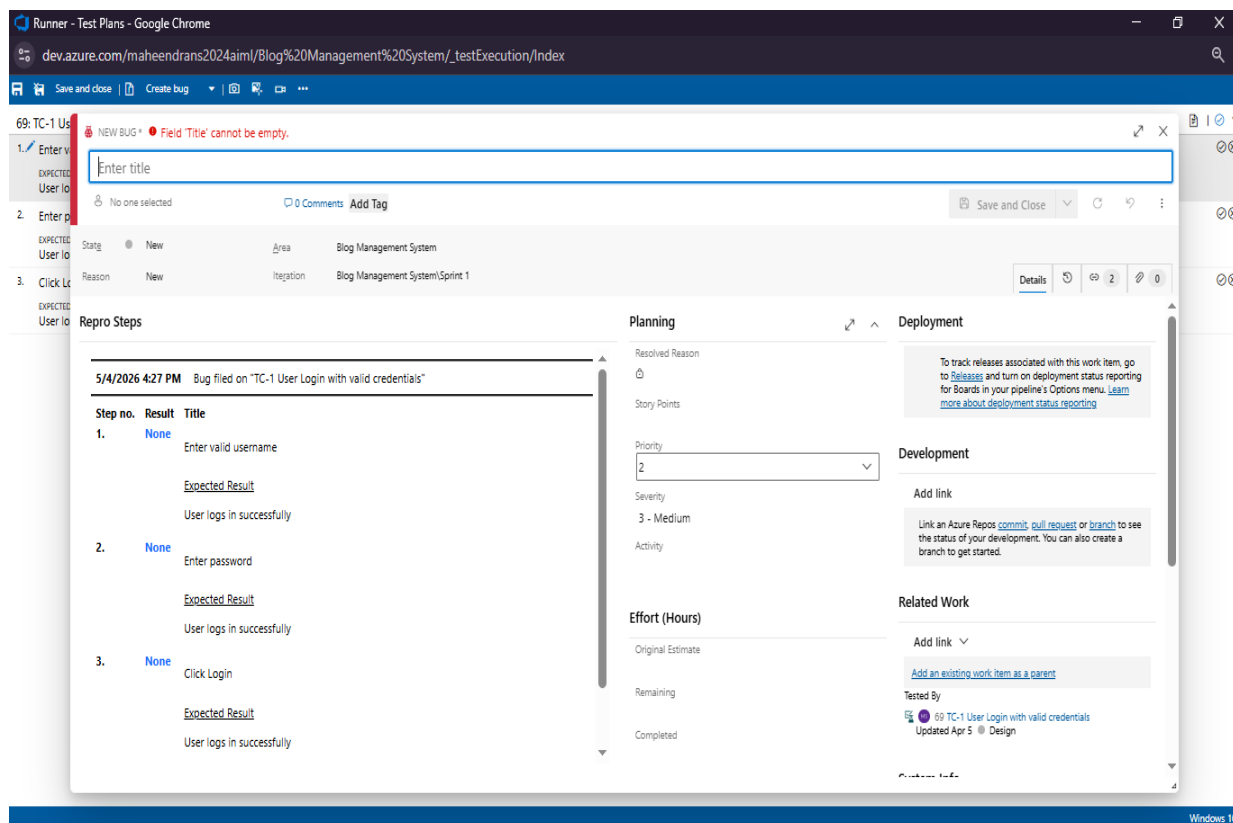
The screenshot shows the 'Runner - Test Plans - Google Chrome' window displaying the execution details of a test case. The URL is 'dev.azure.com/maheendrass2024aiml/Blog%20Management%20System/_testExecution/Index'. The test case is '69: TC-1 User Login with valid credentials'. The execution steps are listed below:

1. Enter valid username
EXPECTED RESULT: User logs in successfully
2. Enter password
EXPECTED RESULT: User logs in successfully
3. Click Login
EXPECTED RESULT: User logs in successfully

4. Recording the test case



5. Creating the bug



Repro Steps

Step no.	Result	Title	Expected Result	Actual Result
1.	Passed	Click 'New Post'	Blog post created successfully	
2.	Passed	Enter title & content	Blog post created successfully	
3.	Failed	Click publish	Blog post created successfully	

System Info

Browser - Name	Google Chrome 104
Browser - Language	en-US
Browser - Height	728
Browser - Width	1280
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36; Chrome/104.0.5112.101; Safari/537.36)
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x64
Operating system - Processor model	Intel(R) Core(TM) i5-10210U CPU @ 2.20GHz
Operating system - Number of processors	4
Memory - Available	16.0 GB
Memory - Capacity	16.0 GB
Display - Pixels per inch (DPI)	96
Display - Width in pixels	1920
Display - Height in pixels	1080

Test Results

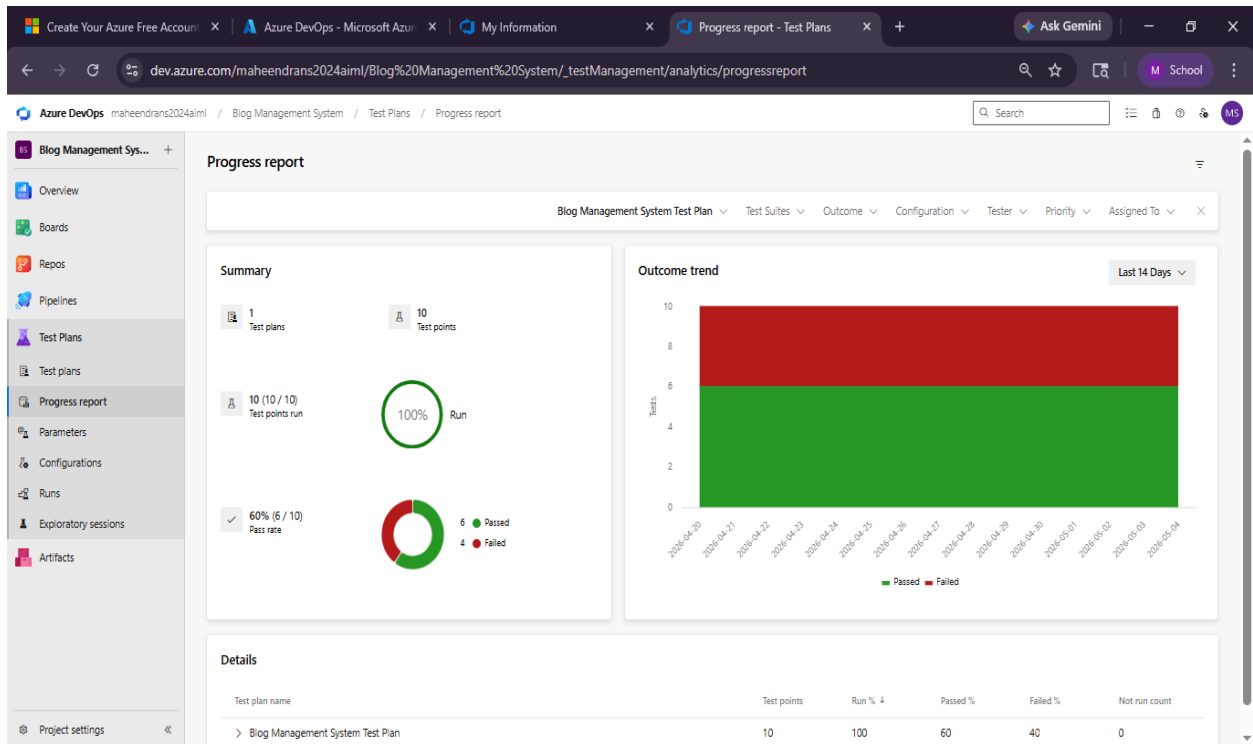
Blog Management System Test Plan (ID: 68)

Define Execute Chart

Test Points (10 items)

Test Point	Outcome
TC-1 User Login with valid credentials	Passed
TC-2 User Login with invalid credentials	Failed
TC-3 Create new blog post	Failed
TC-4 Edit blog post	Passed
TC-5 Delete blog post	Passed
TC-6 View blog post	Failed
TC-7 Add comment to blog	Passed
TC-8 Search blog posts	Passed
TC-9 Logout	Failed
TC-10 Validation for empty blog post	Passed

6. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO : 10	CREATION OF WEB PAGE
16/03/2026	

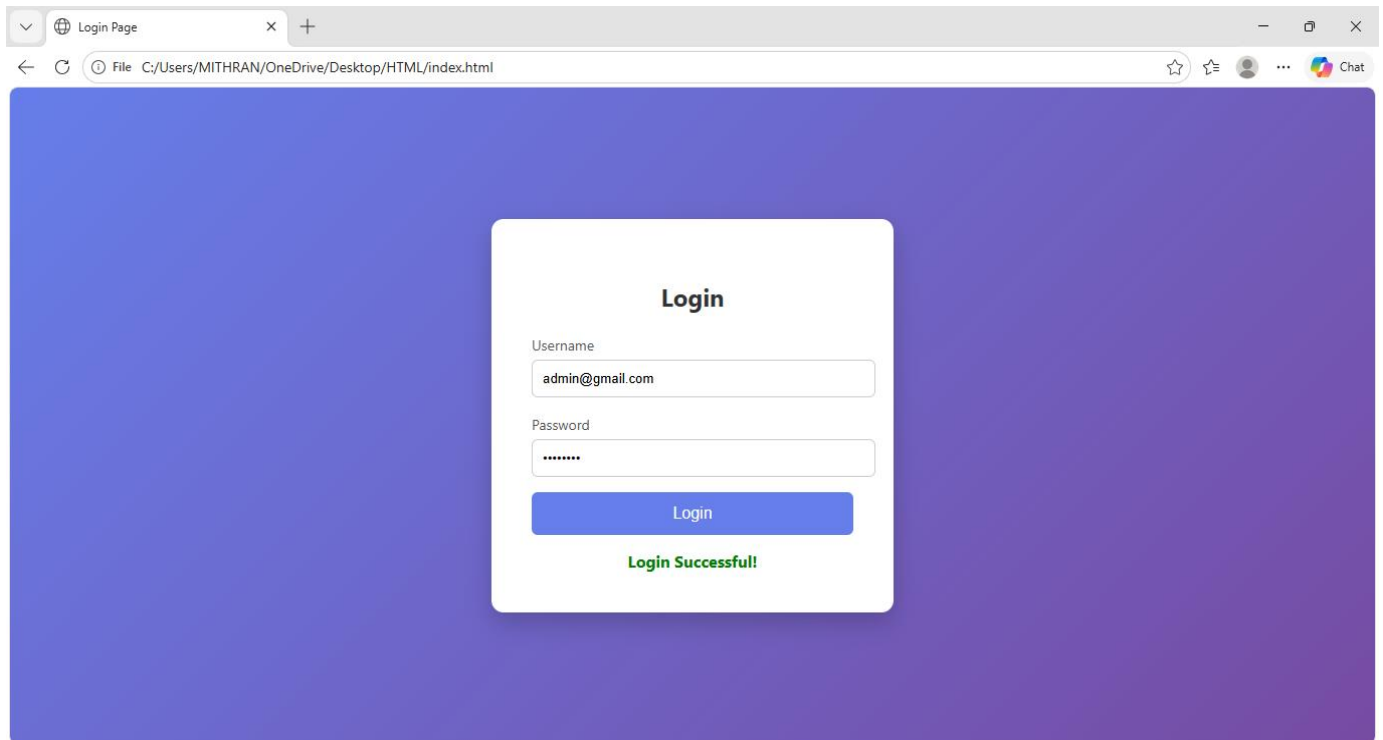
Aim

To design and create a basic web page using HTML (and optionally CSS) to display structured content such as text, images, and links in a web browser.

Algorithm

1. **Start**
2. Open a text editor (e.g., Notepad or VS Code)
3. Create a new file and save it with .html extension
4. Write the basic HTML structure:
 - <html>
 - <head> (title of the page)
 - <body> (content of the page)
5. Add elements such as headings, paragraphs, images, and links
6. Apply basic styling using CSS (optional)
7. Save the file
8. Open the file in a web browser
9. Verify the output and make necessary changes
10. **End**

OUTPUT:



Result

The web page is successfully created and displayed in a web browser with the required content, structure, and basic styling.

EXP NO: 11

06/04/2026

CI/CD PIPELINES IN AZURE

AIM:

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the Student Management System, ensuring faster delivery and improved software quality.

PROCEDURE:

Steps to Create and implement pipelines in Azure:

1. Sign in to Azure DevOps and Navigate to Your Project
Log in to dev.azure.com, select your organization, and open the project where your Student Management System code resides.
2. Connect a Code Repository (Azure Repos or GitHub)
Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.
3. Create a New Pipeline
Go to the Pipelines section on the left panel and click “Create Pipeline”.
Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.
4. Choose the Pipeline Configuration
You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup.
If using YAML, Azure DevOps will suggest a template or allow you to define your own.
5. Define Build Stage (CI - Continuous Integration) from YAML file
6. Install dependencies (e.g., npm install, dotnet restore)
7. Build the application (dotnet build, npm run build)
8. Run unit tests (dotnet test, npm test)
9. Publish build artifacts to be used in the release stage

10. Save and Run the Pipeline for the First Time

Save the YAML or build definition and click “Run”.

Azure will fetch the latest code and execute the defined build and test stages.

11. Configure Continuous Deployment (CD)

Navigate to the Releases tab under Pipelines and click “New Release Pipeline”.

Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).

12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.

13. Set Triggers and Approvals

Enable continuous deployment trigger so the release pipeline runs automatically after a successful build.

For production environments, configure pre-deployment approvals to ensure manual verification before release.

14. Monitor Pipelines and Manage Logs

View all pipeline runs under the Runs section.

Check logs for build/test/deploy stages to debug any errors.

You can also integrate email alerts or Microsoft Teams notifications for build failures.

15. Review and Maintain Pipelines

Regularly update your pipeline tasks or YAML configurations as your application grows.

Ensure pipeline runs are clean and artifacts are stored securely.

Integrate quality gates and code coverage policies to maintain code quality.

Browser tabs: Create Your Azure Free Account, Azure DevOps - Microsoft Azure, My Information, Pipelines - Recent, Ask Gemini, M School.

Address bar: dev.azure.com/maheendrans2024aiml/Blog%20Management%20System/_build

Azure DevOps maheendrans2024aiml / Blog Management System / Pipelines

Left sidebar: Blog Management Sys..., Overview, Boards, Repos, Pipelines, Pipelines, Environments, Library, Test Plans, Artifacts, Project settings

Pipelines

Recent All Runs Filter pipelines

Recently run pipelines

Pipeline	Last run
Maheendran2006.Software-Construction	#20260504.1 • Set up CI with Azure Pipelines Individual CI for main 9h ago 21s

Browser tabs: Create Your Azure Free Account, Azure DevOps - Microsoft Azure, My Information, Pipelines - Runs for Maheendran2024aiml, Ask Gemini, M School.

Address bar: dev.azure.com/maheendrans2024aiml/Blog%20Management%20System/_build?definitionId=1

Azure DevOps maheendrans2024aiml / Blog Management System / Pipelines / Maheendran2006.Software-Construction

Left sidebar: Blog Management Sys..., Overview, Boards, Repos, Pipelines, Pipelines, Environments, Library, Test Plans, Artifacts, Project settings

Maheendran2006.Software-Construction

Runs Stages Branches Analytics Edit Run pipeline

Description	Stages
#20260504.1 • Set up CI with Azure Pipelines Individual CI for main c5b9c9bf	✓

Browser tabs: Create Your Azure Free Account, Azure DevOps - Microsoft Azure, My Information, Pipelines - Run 20260504.1, Ask Gemini.

URL: dev.azure.com/maheendrans2024aiml/Blog%20Management%20System/_build/results?buildId=18&view=results

Azure DevOps maheendrans2024aiml / Blog Management System / Pipelines / Maheendran2006.Software-Construction / 20260504.1

Left sidebar: Blog Management Sys..., Overview, Boards, Repos, Pipelines, Pipelines, Environments, Library, Test Plans, Artifacts, Project settings

#20260504.1 • Set up CI with Azure Pipelines

Maheendran2006.Software-Construction

Run new

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary Code Coverage

Individual CI by Maheendran2006

Repository and version: Maheendran2006/Software-Construction

Time started and elapsed: Today at 12:50 PM, 21s

Related: 0 work items, 0 artifacts

Tests and coverage: Get started

View change Parameters

Stages Jobs

Stage	Jobs
1 job completed 7s	
Job 3s	

Rerun Stage

RESULT:

Thus the pipelines for the given project “Blog Management System has been executed successfully

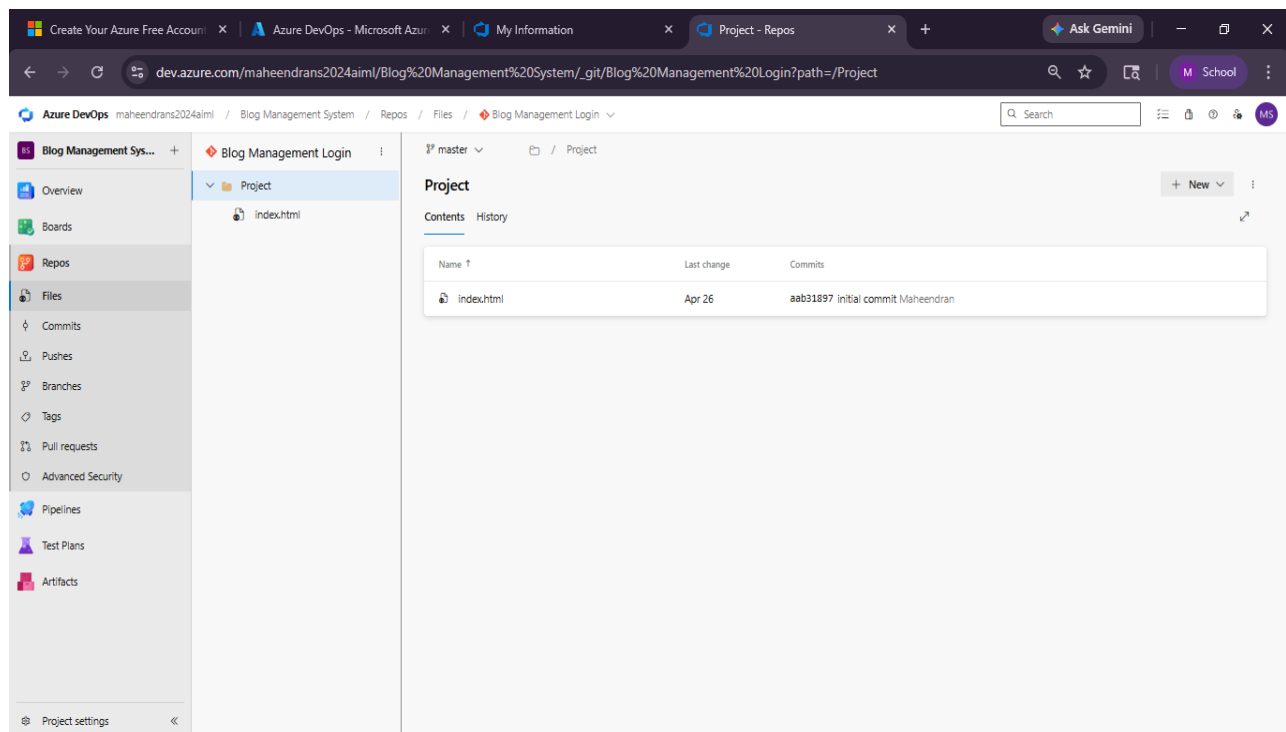
EXP NO: 12
20/04/2026

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Blog Management System project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.